Indian Institute of Information Technology Pune

Certificate Course on Cyber Security and Digital Forensics

Sponsored by:

CRCIDF

Center for Research on Cyber Intelligence and Digital Forensics

Batch-2

1 Month | Hybrid Mode

Dec 4th 2023 to Jan 3rd 2024

Coordinators

Dr. Nitesh Bharadwaj,
Assistant Professor, Dept. of CSE, IIIT Pune

Dr. Bhupendra Singh,
Assistant Professor, Dept. of CSE, IIIT Pune

About IIIT Pune

Indian Institute of Information Technology (IIIT) Pune is committed to providing the best-in-class technical education in the country. The institute is facilitating knowledge transfer while keeping up with the latest development pedagogy. IIIT Pune is an Institute of national importance established by the Ministry of Education (MoE), Government of India, and Industry partners on a Not-for-profit Public Private Partnership (N-PPP) basis. IIIT Pune is functioning from a transit campus located at Survey No. - 9/1/3.Ambegaon Budruk, Sinhagad Institute Road, Pune 411041.The permanent campus for IIIT Pune is over 100 acres at Nanoli village, Tal. Maval, Pune.

IIIT Pune offers undergraduate programs, master degree and a Ph.D. in Electronics and Communications Engineering and Computer Science & Engineering. The academic curriculum for the UG programs is designed to match international standards in engineering education with a view to produce quality engineers. The institute is actively working on revamping its undergraduate curriculum to meet the future IT requirements. Apart from academics, the institute provides a comprehensive environment that promotes art, culture, sports, societal contributions, and self-governance.

Undergraduate students participate in ongoing research and technology development, as a result, a vibrant undergraduate program exists to fulfil the needs of society. The Ph.D. programme is aimed at assisting students acquiring proficiency in their chosen area of research. The academic programme leading to the Ph.D. degree is broad-based and involves a course credit requirement and a research thesis submission. The Institute encourages research in interdisciplinary areas through a system of joint supervision and interdepartmental group activities.

From the Desk of Director

Dr. O. G. Kakde

Indian Institute of Information Technology Pune, established in 2016, is one among the fastest growing IIITs in 20 newly established IIITs by the MOE, Government of India and industry partners as Not-for-profit Public-Private Partnership (N-PPP) Institution. IIIT Pune is committed to impart high-quality technical education and instil a long-term vision amongst the students. Our focus lies not just on academic brilliance but also on the knowledge that is pervasive in areas of emerging technologies. We, through our learning and research programs aim at building such applications of Information Technology whose benefits percolate even to grass root levels.

We not only believe in making our graduates proficient scholars but responsible members of society. Our institute is endowed with state-of-art facilities and has an environment that is conducive to learning and exploration. Every department in the institute has state-of-the-art laboratories. IIIT Pune focuses on interaction with the best institutes, nationally and internationally, through the academic exchange programmes, and also fosters industry-academia partnership among others.

Several Industry-Institute Partnership Programmes, Soft-Skill Trainings, competitions, internships and student projects are organized to cater to communication skills, commitment, and ethics for the comprehensive growth of an individual. The institute promotes innovation and entrepreneurship by encouraging and supporting innovative and novel ideas. Transformation does not merely happen; it requires a plan and a support system. So, while studying at Indian Institute of Information Technology Pune, you will be expected to work hard, have a true and right attitude in the subject area. As you choose to walk on the trail of an exciting and fruitful journey of your life, ensure to walk boldly and diligently.

All the very best!

Dr. O. G. Kakde
Director (IIIT Nagpur)
Additional Charge, Director
IIIT Pune

Course Overview

This course is aimed at training cybersecurity professionals in identifying and reporting the security bugs, especially those related to security exploits and vulnerabilities in a legal and ethical manner. The course allows one to upscale and pursue a career in challenging, adventurous and rewarding field of bug bounty research with hands on experience. The proposed program is an Instructor-led program that covers essential training on cyber security and digital forensic investigation techniques. The course provides knowledge of real-world scenarios with hands-on experience on case studies to develop forensic skills. The course delves into the concepts of forensic investigation, security, volatility framework, autopsy, scalpel, Nmap, DNS pharming, DoS attack, session high jacking, hashcat, etc. that are essential to become an advanced cybersecurity professional. The proposed program is designed to equip students and professionals aspiring to start a technical career in cyber security. This is a technical skilling program, essential to launch a professional career in the exciting domain of Cyber Security and Digital Forensics.

Course Highlights

Learning Outcomes

- Proactively investigate complex security threats, allowing them to investigate, record, and report cybercrimes to prevent future attacks.
- Identify & Evaluate Information Security threats and vulnerabilities in Information Systems and apply security measures to real time scenarios
- Identify common trade-offs and compromises that are made in the design and development process of Information Systems
- Demonstrate the use of standards and cyber laws to enhance information security in the development process and infrastructure protection

Course Curriculum

Module-1: Basics of Cyber Security

- Security Goals: Confidentiality, integrity, and availability,
- Authentication, Authorization and, Auditing.
- Information security risk management.

Module-2: Basics of Networking

- OSI/TCP
- Networking devices and security equipments
- Protocols
- Subnetting and super netting
- Cryptography and hashing

Module-3: Vulnerability Analysis Process, Tools and Techniques

- Active & Passive Foot-printing,
- Protocol Analyzers and Port Scanners,
- OS Fingerprinting,
- Vulnerability Scanners.

Module-4: Malware and Static & Dynamic Analysis

- Social engineering
- Malicious logic, Trojan horses, viruses.
- Boot sector and executable infectors.
- Spyware, adware, and ransomware.

Module-5: Securing Linux System: Access Control mechanism

- Discretionary access control
- Mandatory access control
- Role-based access control
- Linux permission model
- Shadow and password file

Module-6: Digital Forensics: Process, Tools & Techniques and Research Challenges

- Definition of Digital Forensics
- Disk Drives and their characteristics
- Understanding Hard Disk Partitions
- Booting Process in different Operating Systems
- Slack Space
- Forensics Imaging
- Imaging using FTK Imager

Module-7: Understanding Hard Disks and File Systems (NTFS & Ext4)

- Disk Drives and their characteristics
- Understanding Hard Disk Partitions Booting Process in different Operating Systems
- Slack Space
- Metadata in NTFS and Ext4 file systems

Module-8: Data Acquisition: Imaging and Cloning

- Hashing and Write Blockers
- Forensics Imaging in Kali Linux using dd, dcfldd, dc3dd
- Imaging using FTK Imager

Module-9: File Systems Analysis using TSK

- Analyzing disk images using TSK utilities

Module-10: Windows Forensics

- Registry Forensics
- Recycle Bin Forensics
- Jump List Forensics
- SRUDB.dat forensics
- Prefetching in Windows
- Program Executions Artifacts

Module-11: Anti-forensics and Anti-anti-forensics

- Data Hiding into Slack Space
- Secure Deletion
- Timestomping

Module-12: Volatile Memory Forensics

- Why/What Memory Forensics
- Volatility Configuration
- Volatility Analysis
- Windows Memory Analysis

Module-13: Linux Forensics

- Basic Linux Commands
- File Hierarchy Standard
- Hunt Users and Groups
- File Hunting
- Failed logins and Actors IP address
- Timestamps and Deleted files in Journal

Module-14: Email and Drone Forensics

- Examining Email messages
- Email Server Examination
- Tracing emails
- Email Forensics Tools
- Introduction to Drone forensics and challenges

Module-15: Password Recovery

- Password Cracking Methods
- Password Cracking Tools
- Hashcat for Windows password cracking

Module-16: Mobile Forensics

- Phone Locks and Discuss Rooting of Android
- Mobile Forensics Process
- Perform Logical Acquisition on Android
- Perform Physical Acquisition on Android
- Mobile Forensics Challenges
- CDR Analysis

Tools and Frameworks

Eligibility Criteria

- Diploma/Graduate/Post Graduate from Institutes/Universities recognized by Association of Indian Universities
- Central/State Govt officials/Employees

Batch Size: Minimum 50 and Maximum 100 participants.

Instruction Mode: **Hybrid** (Participants who wish to attend classes offline can attend, in IIIT Pune. Else, candidate can join the class online via Google Meet. However, we encourage participation in OFFLINE mode for better learning outcomes.)

Who Can Attend?

- UG/PG graduated/pursuing students.
- Ph.D. Scholars from higher education institutes.
- Faculties / Staffs / Lab Instructors from technical and academic institutions.
- Industry/working professionals seeking cyber security skills.
- Any other Cyber Security enthusiasts.

Schedule

Week	Day	10:00 AM to 11:00 AM	11:00 AM to 12:00 PM	12:00 PM to 01:00 PM	01:00 PM to 02:00 PM	02:00 PM to 03:00 PM	03:00 PM to 04:00 PM	04:00 PM to 05:00 PM	05:00 PM to 06:00 PM
Week-1	**Monday**	Lecture-1	Lecture-2	Lecture-3	LUNCH	Practical-1	Practical-2	Practical-3	-
	Tuesday	Lecture-4	Lecture-5	Lecture-6		Practical-4	Practical-5	Practical-6	-
	Wednesday	Lecture-7	Lecture-8	Lecture-9		Practical-7	Practical-8	Practical-9	-
	Thursday	Lecture-10	Lecture-11	Lecture-12		Practical-10	Practical-11	Practical-12	-
	Friday	Lecture-13	Lecture-14	Lecture-15		Practical-13	Practical-14	**Assessment-1**	
Week-2	**Monday**	Lecture-16	Lecture-17	Lecture-18	LUNCH	Practical-15	Practical-16	Practical-17	-
	Tuesday	Lecture-19	Lecture-20	Lecture-21		Practical-18	Practical-19	Practical-20	-
	Wednesday	Lecture-22	Lecture-23	Lecture-24		Practical-21	Practical-22	Practical-23	-
	Thursday	Lecture-25	Lecture-26	Lecture-27		Practical-24	Practical-25	Practical-26	-
	Friday	Lecture-28	Lecture-29	Lecture-30		Practical-27	Practical-28	**Assessment-2**	
Week-3	**Monday**	Lecture-31	Lecture-32	Lecture-33	LUNCH	Practical-29	Practical-30	Practical-31	-
	Tuesday	Lecture-34	Lecture-35	Lecture-36		Practical-32	Practical-33	Practical-34	-
	Wednesday	Lecture-37	Lecture-38	Lecture-39		Practical-35	Practical-36	Practical-37	-
	Thursday	Lecture-40	Lecture-41	Lecture-42		Practical-38	Practical-39	Practical-40	-
	Friday	Lecture-43	Lecture-44	Lecture-45		Practical-41	Practical-42	**Assessment-3**	
Week-4	**Monday**	Lecture-46	Lecture-47	Lecture-48	LUNCH	Practical-43	Practical-44	Practical-45	
	Tuesday	Lecture-49	Lecture-50	Lecture-51		Practical-46	Practical-47	Practical-48	
	Wednesday	Discussions and Doubt Clearance				Practical-49	Practical-50	Practical-51	
	Thursday	Theory Evaluation				Practical Evaluation			
	Friday	Report Submission*				Certificate Distribution and Valedictory Session			

*2 Page report submission in the last session on the learning outcomes.

Attendance, Assessment and Evaluation

- Minimum required 80% attendance to receive a certificate.
- Minimum 70% score is required in both MCQ test and practical evaluation.
- Mandatory three assessments, one after each week
- (Combination of MCQs/short answer type/scenario based/assignments.)
- Schedule of final evaluation and certificate distribution.

Week-4	**10:00 AM to 01:00PM**	**02:00 PM to 05:00PM**
Thursday	Theory Evaluation	Practical Evaluation
Friday	Report Submission*	Certificate Distribution and Valedictory Session

***2 Page report submission in the last session on the learning outcomes.**

Certification

Participants who successfully meet the evaluation criteria and satisfy the requisite attendance criteria, will be awarded a `Certificate of Completion'.

Registration Link: https://forms.gle/HBZv9LXKJ27yFBzb6

Course Coordinators

Dr. Nitesh K Bharadwaj

Dr. Nitesh K Bharadwaj is serving as an Assistant Professor in the Department of Computer Science & Engineering, IIIT Pune. His area of interest includes digital forensics, storage drive analysis, drone forensics, VLSI design, Internet of Things, and Cyber Security Technologies. His research combines the mathematical concepts in the field of digital forensics to support digital forensics community in efficient examination of voluminous data. He is involved in providing innovative solution to real world problem and one of his contributions during pandemic to define social safety circle using augmented reality was applauded by Honorable Minister of Education in the year 2020. Before developing and managing the course on Cyber Security and Digital Forensics, he taught Cyber Security and Digital Forensics, C-programming, Data Structures, Advanced Web Technology, Advanced data Structures, Computer Networks, Internet of Things, Computer Organization and Architecture to the undergraduate students. Bharadwaj hold Ph.D. in Computer Science & Engineering from Defence Institute of Advanced Technology (DIAT) Pune and M.Tech. from ABV-Indian Institute of Information Technology & Management (IIITM) Gwalior in Computer Science and Engineering (Specialization VLSI Design). He completed his undergraduate B.E. in Computer Science & Engineering from CSVTU Bhilai.

Dr. Bhupendra Singh

Dr. Bhupendra Singh is serving as an Assistant Professor in the Department of Computer Science & Engineering, IIIT Pune. His area of interest includes digital forensics, Windows forensics, database forensics, and Cyber Security Technologies. His research contribution involves development of custom tools for various Operating Systems and metadata forensics in the field of digital forensics for crime investigation. He has published several research articles in peer-reviewed journals and book chapters. He serves on the Technical Program Committee member of many international conferences on Digital Forensics and Cyber Security. He has taught Cyber Security and Digital Forensics, Python programming, Cloud Computing, to the undergraduate students. Singh hold Ph.D. in Computer Science & Engineering from Defence Institute of Advanced Technology (DIAT) Pune and M.Tech. from Central University of Rajasthan in Computer Science and Engineering.

Course Fee

S. No.	Particulars	Fees*
1	**Course Fee**	**INR 29,500 (including 18% GST)**
2	**Accommodation***	**Guest Charge**
3	**Mess**	**On Payment Basis**

Note: *Subject to the availability in the IIIT Pune Hostel. Accommodation & Mess charges are applicable only for participants attending in offline mode.
The **tentative** hostel charge is **INR 5,100** including taxes.

Collection of all fees will be done by IIIT Pune, the bank details are as follows:

Bank details for fee submission	
Account Name	**Indian Institute of Information Technology Pune**
Account Number	**39576958196**
Bank Name	**State Bank of India**
Branch Name & Code	**SBI, Ambegaon, Pune (11648)**
IFSC Code	**SBIN0011648**

During campus visit, participants who want to opt the accommodation and mess facilities of IIIT Pune will have to pay the charges directly to IIIT Pune. This shall be subject to accommodation availability at the campus.

Course Timelines (Batch-2)

Application Start Date	8th Aug 2023
Application Closure Date	**31st Oct, 2023**
Course Commencement	**4th Dec, 2023**
Course End Date	**3rd Jan, 2024**

Note: Indian Institute of Information Technology (IIIT) Pune reserves the right to restructure the course content, format, number of sessions, certificate format, terms in the course or can incorporate any such change deemed necessary by the institute without prior intimation.

Course Coordinators

Dr. Nitesh K Bharadwaj

Dr. Bhupendra Singh

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